

Bhavyashree Putta

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Professional Summary

Software Engineer with 2+ years of experience building scalable backend systems, AI-powered applications, and data-intensive platforms processing millions of records. Experienced in designing cloud-native services, REST APIs, and intelligent search systems with a focus on performance, reliability, and scalable software design.

Experience

United Way of Northern New Jersey | Software Engineer Dec 2025 - Present

- Designed and developed **Java Spring Boot microservices, REST APIs, and user-facing features** powering the **United For ALICE** platform, enabling research, reporting, and data exploration across **44 U.S. states**.
- Engineered a **Kafka-based ingestion pipeline** that decoupled Census Bureau and state-partner data producers from downstream services through event replay, **reducing financial hardship calculation time by 45%**.
- Architected **asynchronous report generation** using **RabbitMQ task queues**, generating **40K+ report permutations** across geographic and demographic dimensions while **eliminating blocking API requests**.
- Implemented **Hibernate JPA mappings** and **optimized JPQL/SQL queries**, reducing query latency by **35%** across **20M+ records** for interactive reporting and map-based data exploration.
- Improved production reliability** by implementing **structured logging, error handling, and cross-layer debugging** across backend services, **maintaining 99.9% uptime** and reducing recurring production issues.
- Built a **RAG-powered conversational search assistant** with **JPMorgan Chase**, enabling natural-language querying across 10,000+ research reports and datasets and **reducing information retrieval time by 60%**.

UIC College of Applied Health Sciences | Research Software Engineer June 2025 - Nov 2025

- Built a **Google funded FastAPI RAG chatbot** integrating **Azure OpenAI's GPT-4** with hybrid Pinecone and FAISS retrieval, **achieving 1.2s median latency** for accurate, reliable responses in low-connectivity environments.
- Engineered a **two-tier session architecture** using **Cloudflare D1** for persistent chat history and **KV with TTL-based expiry** for caching, cutting D1 read load and **enabling sub-second access for 50+ concurrent users**.
- Containerized microservices using **Docker**, automated **IaC with Terraform**, and managed **deployments through Kubernetes + Helm**, achieving **60% fewer response failures** and improved observability across pods.

University of Illinois Chicago | Graduate Assistant - Full Stack Developer Aug 2023 - May 2025

- Developed a full-stack transit platform serving 65K+ users, **improving application responsiveness by 45%** through **asynchronous data loading and backend API optimization**.
- Built **Java-based RESTful service** with asynchronous workflows and schema validation, integrating queue-based processing with Azure Service Bus and automated **JUnit test suites to reduce integration failures by 40%**.
- Designed and optimized **relational schemas, indexing strategies, and ACID-compliant transactions**, improving query performance by 60% and enabling zero-downtime releases.
- Integrated **200+ Jest, React Testing Library, and Cypress tests** into CI/CD pipelines, **increasing code coverage by 40%**, reducing escaped defects by 35%, and supporting stable weekly releases.

Projects

GetLect - Full-Stack Recruitment Platform | React, Java, Spring Boot, MySQL, AWS

- Built and deployed a **production recruitment platform** on **AWS EC2**, designing **Spring Boot REST APIs** with a **normalized relational schema** for user onboarding, job postings, and application workflows.
- Implemented **JWT authentication** with **Spring Security** and **paginated, DTO-based endpoints**, **optimizing MySQL performance** through **composite indexing and query tuning** to improve response times by roughly 40%.

Distributed Replication & Consistency Simulator | C++, Python, gRPC, Raft Consensus, Fault Injection

- Built a **distributed key-value store** in C++ using the **Raft consensus algorithm**, ensuring linearizable reads and writes through leader election and **replicated state machines across a 5-node cluster**.
- Developed a **Python fault-injection framework** to simulate **network partitions, message delays, and node failures**, identifying and **resolving split-brain scenarios** during leader re-election.

Technical Skills

Languages: Java, Python, C++, JavaScript (ES6+), TypeScript, SQL

Backend & Distributed Systems: Spring Boot, Hibernate JPA, REST APIs, gRPC, Kafka, RabbitMQ, Microservices

Cloud & DevOps: AWS (EC2, S3, RDS), Azure, Docker, Kubernetes, Terraform, OpenTelemetry, Prometheus, Grafana

Databases: MySQL, Microsoft SQL Server, PostgreSQL, Redis, Elasticsearch, DynamoDB

AI & ML: RAG, Semantic Search, Vector Search, Pinecone, PyTorch, TensorFlow, LangChain, NLP

Education

University of Illinois Chicago, USA
Master of Science in Computer Science

Aug 2023 – May 2025
GPA 3.88